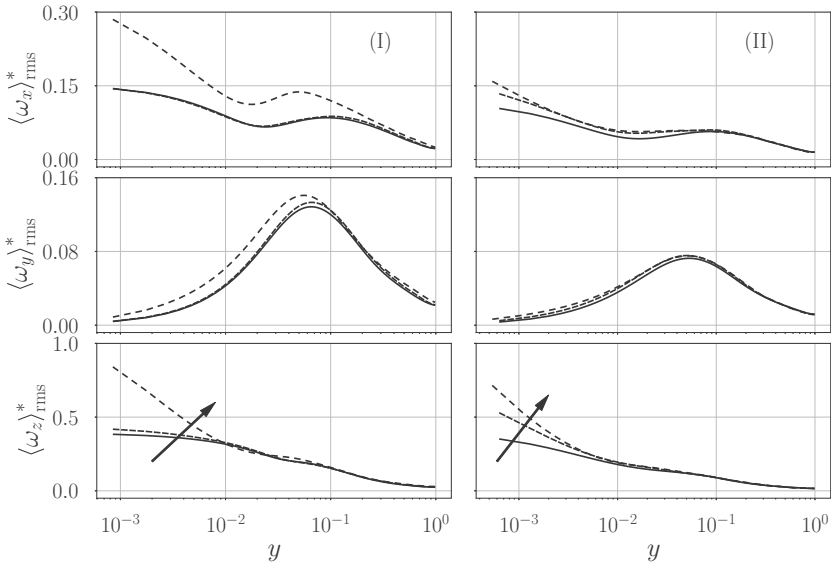

FIG. 4. Root-mean-square fluctuation vorticity components, normalized by $\mu_{\rm ref}$ and the wall shear stress τ_w in the impermeable wall cases. The line spacing in the plot increases with an increasing permeability (decreasing resistance R), also shown by the arrow.

vortex centers [37]. Due to the no-slip condition in u and w, a maximum in $\langle\omega_x\rangle^*_{\rm rms}$ is observed on the surface. The combined effect of streaks and no-slip conditions leads to a local minimum that is roughly at the edge of the pictured vortex structures. After applying the permeable wall condition, significant enhancement of $\langle\omega_x\rangle^*_{\rm rms}$ is observed below $y = 0.2$ and the locations of the local maximum and minimum are shifted closer to the wall. The phenomenon of increasing $\langle\omega_x\rangle^*_{\rm rms}$ is due to the entrainment effect of the wave structures as presented in Fig. 3, which is in fact a three-dimensional event as sketched in Fig. 5. While the high momentum fluid is entrained toward the wall, it is also found to be expelled to the cross direction. This leads to secondary flows that increase the component $\partial w'/\partial y$ and thus $\langle\omega_x\rangle^*_{\rm rms}$. A cross-plane view of the flow field is shown in

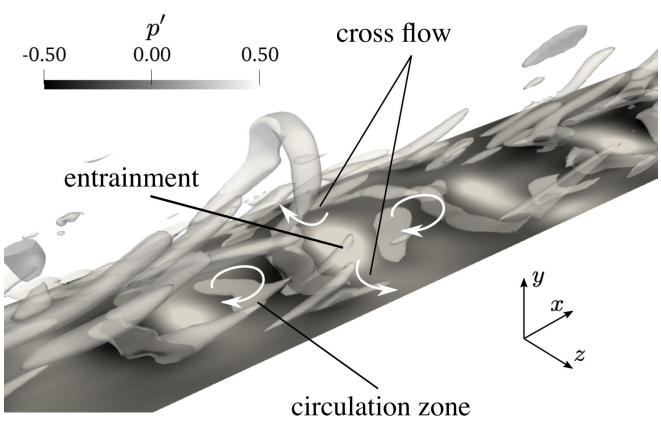

FIG. 5. A sketch of the flow pattern associated to wave-like structures showing the circulation zone, entrainment region and the resulting cross flow. The background isosurface plot is generated with the case $M_b = 3.50, R = 0.50$).

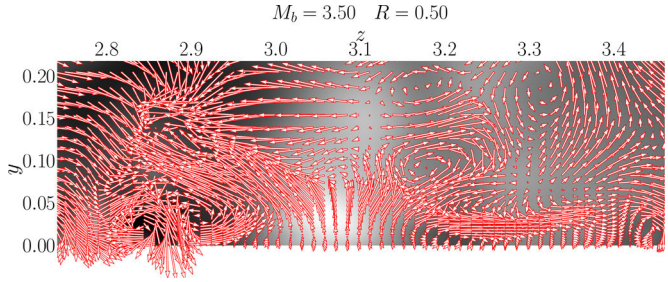

FIG. 6. Instantaneous flow field in the cross-plane (z-y plane) at the location of entrainment (positive pressure fluctuation) on top of the contour plot of pressure fluctuations, taken from the case $M_b = 3.50$, $R = 0.50$. The vector length is proportional to the in-plane velocity magnitude.

Fig. 6 for $M_b = 3.50$, $R = 0.50$, which clearly shows the cross flow on sides of the positive pressure fluctuation blob, where the flow entrainment is present. The effect of permeable walls on $\langle\omega\rangle^*_{z,\mathrm{rms}}$ is mainly confined below $y = 0.02$. Significance enhancement of the strength can be observed below $y = 0.01$, which tends to be the effect of circulation zones captured by Q criterion as spanwise "rollers" as previously shown in Fig. 3.

VI. TKE BUDGET ANALYSIS

In this section, parts of the budgets of turbulence kinetic energy (TKE) will be examined. The transport equation for TKE in compressible flows, including the SGS contribution, is given by

$$\overline{\rho}\frac{\partial k}{\partial t} + \overline{\rho}\widetilde{u}_j\frac{\partial k}{\partial x_j} = P^k - \epsilon + D^k + T^k + D^k_p + W^k + \Pi^k_p - \epsilon_{\mathrm{sgs}} + D^k_{\mathrm{sgs}}, \tag{30}$$

where $k = \frac{1}{2}\widetilde{u''_i u''_i}$ represents the TKE. Two groups of the terms on the right-hand side of TKE budget are selected for further examination:

(1) Group I: Production and dissipation

$$P^k = -\overline{\rho u''_i u''_j}\frac{\partial \widetilde{u}_i}{\partial x_j}, \quad \text{production}, \tag{31a}$$

$$\epsilon = \overline{\tau_{ji}\frac{\partial u''_i}{\partial x_j}}, \quad \text{resolved dissipation}, \tag{31b}$$

$$\epsilon_{\mathrm{sgs}} = \overline{\tau_{ji,\mathrm{sgs}}\frac{\partial u''_i}{\partial x_j}}, \quad \text{SGS dissipation}. \tag{31c}$$

(2) Group II: Pressure-related terms

$$D^k_p = -\frac{\partial \overline{p' u''_j}}{\partial x_j}, \quad \text{pressure diffusion}, \tag{32a}$$

$$W^k = -\overline{u''_i}\frac{\partial P}{\partial x_i}, \quad \text{pressure work}, \tag{32b}$$

$$\Pi^k_p = \overline{p'\frac{\partial u''_i}{\partial x_i}}, \quad \text{pressure dilatation}. \tag{32c}$$

Unless otherwise mentioned, all budget terms are normalized by the quantity $\tau_w U_b/\delta$ taken from the present impermeable-wall simulations. Some available reference data for impermeable-wall calculations [31] will also be included.

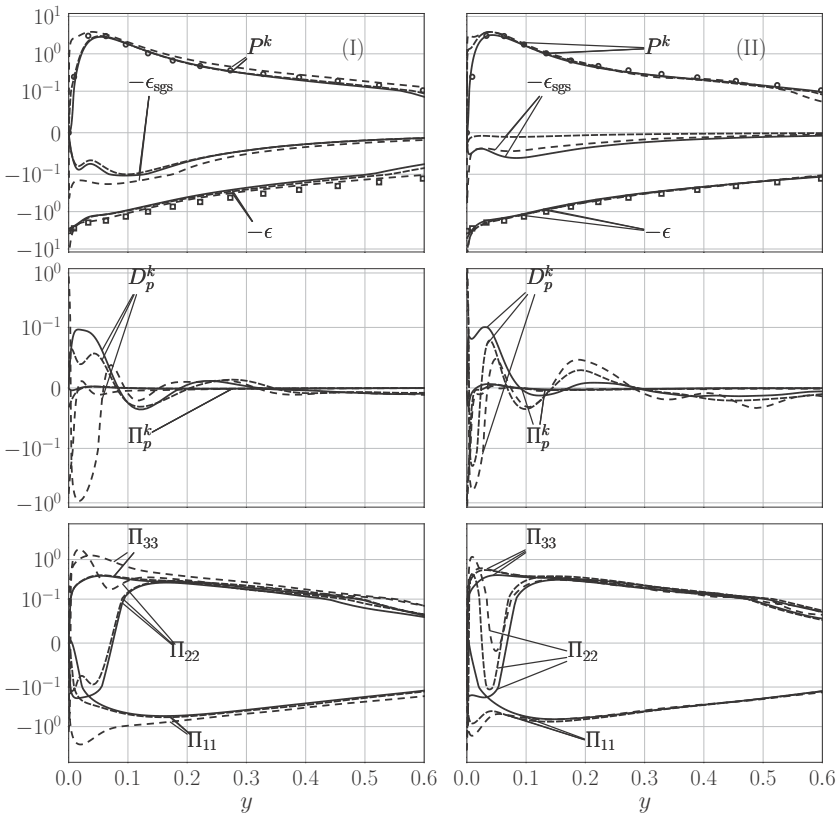

FIG. 7. Budgets terms below $y = 0.6$ for all simulations. Plots in the same column belong to the same Mach, indicated by labels "(I)" and "(II)." First row: Turbulence production, P^k, resolved dissipation, ϵ, and SGS dissipation, ϵ_{sgs}. Second row: pressure diffusion D_p^k and pressure dilatation Π_p^k. Third row: pressure-strain terms $\Pi_{ii}, \quad i = 1, 2, 3$ as will be given in Eq. (37). The line spacing in the plot increases with an increasing permeability (decreasing resistance R). For the same Mach number, quantities in cases of various resistance R are normalized by values taken from impermeable wall simulations.

A. Group I: Production and dissipation

In this section the turbulent production P^k and resolved dissipation ϵ are examined. For completeness, the SGS dissipation ϵ_{sgs} is also included. All these terms are plotted in Fig. 7. The dissipation term can be further decomposed into a solenoidal part, ϵ_s, and a dilatational part, ϵ_d, where

$$\overline{\rho}\epsilon_s = \overline{\nu}\overline{\rho\omega_i''\omega_i''}, \quad \overline{\rho}\epsilon_d = \tfrac{4}{3}\overline{\nu}\overline{\rho u_{i,i}'' u_{i,i}''}. \tag{33}$$

Their ratio, ϵ_d/ϵ_s, is (surprisingly) found to be negligible for all cases and is not included in Fig. 7. As expected, cases with $R = 1.00$ exhibit no significant differences in the budgets as compared to the baseline impermeable cases due to their relatively low permeability (or high resistance R). Effects on the budgets due to the permeability start to show in lower resistance cases with $R = 0.50$. Higher turbulent production is observed in the near-wall region, with peaks of production shifted closer to the wall; the fluid layer of high turbulent production also widens. The change in turbulent production agrees with the trend shown in the mean velocity and Reynolds-shear stress previously shown in Fig. 2. Exploiting the statistical homogeneity in the x and z directions the production term

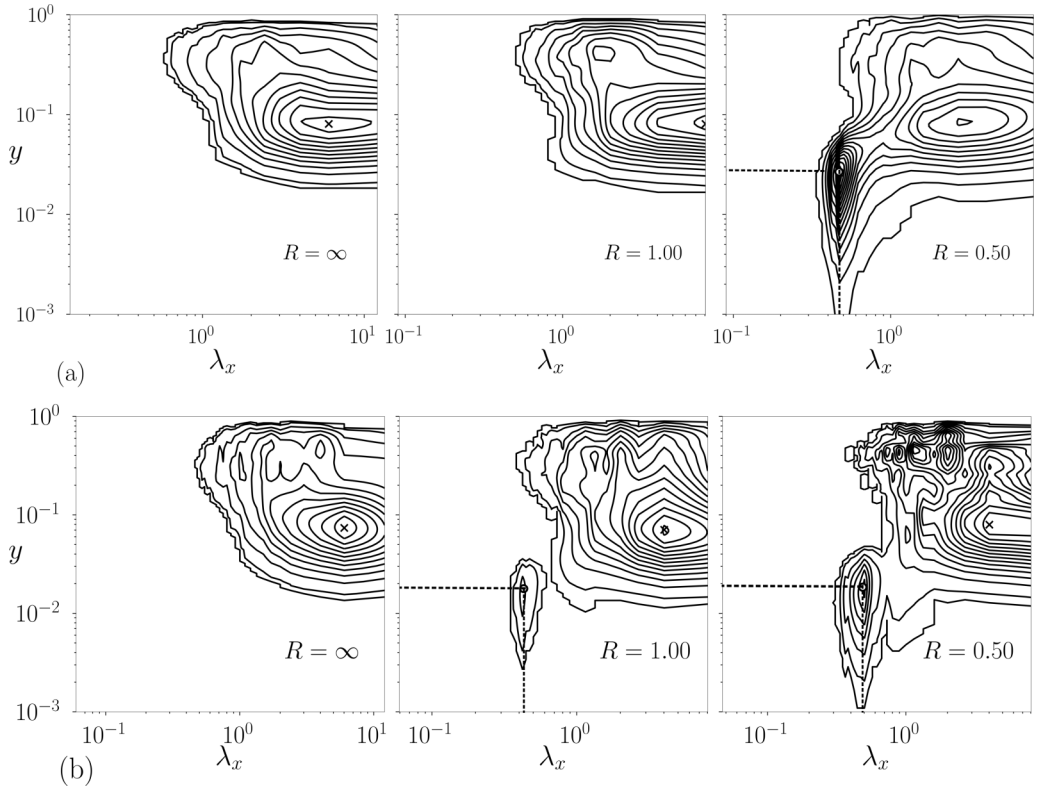

FIG. 8. Premultiplied production spectrum as a function of distance from the wall and streamwise wavelength at $M_b = 1.50$ (a) and $M_b = 3.50$ (b). The "x" symbols indicate the highest contribution to the production spectra. Dashed lines indicate the location and the wavelength of the contribution associated to the waves due to porous walls. The impermeable cases ($R = \infty$) have longer domains, hence the larger maximum λ_x.

reads

$$P^k = -\overline{\rho u'' v''}\frac{d\widetilde{u}}{dy}. \tag{34}$$

The reaction of the impedance wall to the overlying turbulence enhances the near-wall momentum transport, leading to higher wall shear (thus, high $d\widetilde{u}/dy$) and the Reynolds shear stress, which eventually results, in turn, in an enhancement of the turbulent production. Permeability also leads to a spike in dissipation, with a ten-fold increase at the wall as compared to impermeable baselines.

The observed changes in the TKE budgets are due to the instability wave pattern shown in Sec. V, which has been confirmed in the companion work [16] by showing the contribution of TKE production terms of different wavenumbers at different locations, though for slightly different wall permeabilities for each Mach number. Here the same methodology applied to the cases in this work, as shown in Fig. 8. The production spectrum reads

$$\widehat{P}^k = -\frac{1}{2}\left[\left\langle \Re\left\{\widehat{\rho u'}\widehat{v'}^\star \frac{d\widetilde{u}}{dy}\right\}\right\rangle_{z,t} + \left\langle \Re\left\{\widehat{\rho v'}\widehat{u'}^\star \frac{d\widetilde{u}}{dy}\right\}\right\rangle_{z,t}\right], \tag{35}$$

where the superscript * represents the complex conjugate of a quantity, and $\langle\cdot\rangle_{z,t}$ indicates an averaging operator in spanwise direction and time, and $\widehat{\cdot}$ the Fourier transform operator. This is similar to the method used by Ref. [38], with a modification to account for compressibility. It is also found that there is no significant difference between the two terms on the right-hand side of

Eq. (35) and thus an average operation is performed. Figure 8 shows the contour line plot of the premultiplied spectra

$$\widehat{\Phi}^*_{P^k,x} = \mathrm{Re}^*_\tau \kappa_x y \widehat{P}^k \mu / \tau^2_{w,R=\infty} \tag{36}$$

as a function of streamwise wavelength and distance from the wall, normalized by the shear stress of the reference impermeable cases.

At the lower Mach number with $R \geqslant 1.00$, most of the production is due to wavelengths $\lambda_x \geqslant 2.0$ and is concentrated in the region $y \in (0.05, 0.2)$, which roughly corresponds to $y^* \in (15, 50)$ of the impermeable wall case. As the resistance decreases to $R = 0.50$, the spectra exhibit the same pattern in the same region. However, a second region of high contribution has appeared in the range $y \in (0.01, 0.05)$, whose wavelength matches the length scale of the wave structure shown in Fig. 3. For $M_b = 3.50$, a similar behavior is observed and the second region also appears for the higher resistance case ($R = 1.00$). In either Mach number, the wave pattern of the instability observed in the flow is responsible for the enhancement in the near-wall production.

B. Group II: Pressure-related terms

All pressure-related terms in the TKE energy budgets are presented in this section, including pressure diffusion D^k_p and pressure dilatation Π^k_p. The term W^k_p, i.e., pressure work, is found to be negligible across the whole channel even with high wall permeability and it thus omitted from the plots. All these three terms are known to be small as compared to other terms in an impermeable wall channel flow. The pressure diffusion term D^k_p characterizes the TKE diffusion process caused by the fluctuating pressure, which tends to reduce the spatial heterogeneity in the TKE distribution. For an impermeable wall, a local maximum with a normalized value around 0.1 is observed near the wall at $y \approx 0.03$, serving as a gain term in the TKE budget, that is, TKE is transported into this region. The magnitude of D^k_p becomes negligible above $y = 0.2$. For permeable wall cases with $R = 1.0$ at $M_b = 1.50$, attenuation of the transport in the same region is observed, with no significant changes in other part of channel except for a nonzero value at the porous wall. At higher Mach number $M_b = 3.50$ for the same resistance, the peak changes sign and becomes negative, indicating TKE being diffused away from this region. Same negative peaks are observed at lower resistances $R = 0.50$ for both Mach numbers. In addition, a value around 1.0 is achieved on the wall surface as compared to zero for the impermeable wall. This behavior indicates that for the cases with the highest wall permeability, TKE is being diffused toward the porous surface by the pressure fluctuations. The value of pressure dilatation Π^k_p is negligible across the whole channel over an impermeable wall. The presence of an impedance wall merely results in a spike of negative pressure dilatation in the region $y \leqslant 0.02$. As shown in previous sections, a redistribution of energy in the normal components of Reynolds stresses is observed at both Mach numbers. It is therefore interesting to examine the pressure-strain term that is known to be responsible for TKE redistribution [33]. The pressure-strain term in normal components of Reynolds stresses can be obtained by deriving the corresponding transport equation given in Appendix D. The pressure-strain term then reads

$$\Pi_{ii} = \overline{2p' \frac{\partial u''_i}{\partial x_i}}, \quad i = 1, 2, 3. \tag{37}$$

Note that the summation over repeated indices is not used here, and i represents ith coordinate direction. The above terms are plotted in Fig. 7 for all cases presented. Π_{11} acts as a sink in the budget of $\widetilde{u''u''}$ across the whole domain, with a maximum peak achieved around $y = 0.08$ over an impermeable wall. The effect of the permeability is to further enhance the sink effect below $y = 0.2$, mainly for cases with $R = 0.50$ and permeable cases at $M_b = 3.50$. However for each Reynolds normal stress, it is the balance of its production, dissipation (components) and the pressure-strain that eventually determines the intensity, which is the reason that $\widetilde{u''u''}$ show slightly different behavior for different Mach numbers with resistance $R = 0.50$. The production of Π_{22} and Π_{33} are zero due to statistical homogeneity. For the rest two normal stresses, the production terms

are zero due to double statistical homogeneity. Permeability attenuates the sink effect of Π_{22} and enhance the source effect of Π_{33} in regions close to the wall, both of which lead to an increase in the respective normal stresses. This agrees with the observed redistribution of normal stresses in Fig. 2.

VII. CONCLUSION

To investigate the effect of porous media on supersonic turbulent flow field, large-eddy simulations of compressible turbulent channel flows over complex wall impedance at two bulk Mach numbers $M_b = 1.50, 3.50$ were carried out. The bulk Reynolds numbers of cases presented are adjusted to keep the viscous Reynolds number fixed around $\mathrm{Re}_\tau^* = 220$. A time-domain impedance boundary condition (TDIBC) based on auxiliary differential equation (ADE) method is applied to bottom wall of the channel, using a three-parameter model with a resonating frequency tuned to most energetic frequency in the turbulent channel flow. Two values of resistance $R = \{0.50, 1.00\}$, which is inversely proportional to the wall permeability at resonating frequency, are chosen for each Mach number. Results show that finite wall permeability has a significant effect on the near-wall turbulence with sufficiently low resistance value $R = 0.5$, while cases with $R = 1.0$ show no appreciable difference with the impermeable baseline cases. The typical law-of-wall is no longer followed with nonzero wall permeability and the deviation increases as the permeability increases. As compared to impermeable wall cases, higher wall-normal gradient of mean streamwise velocity is observed in permeable wall cases, indicating an increase in the mean wall shear stress. The phenomenon is due to the oscillatory wall normal velocity enhancing turbulent mixing close to the wall. Redistribution of Reynolds normal stresses are shown, with an increase in the components $\widetilde{v''v''}$ and $\widetilde{w''w''}$, and a drop in $\widetilde{u''u''}$ specifically to the case $M_b = 1.50, R = 0.50$, as a result of balancing among production, dissipation and pressure-strain terms.

The change in the mean flow statistics are directly related to the structural changes in the near-wall turbulence. Streamwise-traveling waves with a wavelength $\lambda_x \approx 0.47$ are captured by Q criterion in permeable wall cases, creating local circulations zones separated by entrainment that is directing the flow toward the wall. The analysis of TKE budgets shows that most of the effects are confined near the wall. The turbulence production is significantly enhanced below $y = 0.1$, with the most of contribution in this region coming from the aforementioned wave structures. The dissipation is also increased but its effect is mostly discernible on the wall surface. No obvious changes have been observed in dilatational dissipation. For other budget terms, pressure diffusion is the only one that is obviously altered: TKE is being diffused from where production reaches its maximum toward the wall at $R = 0.50$, opposite to the behavior of low permeability cases. It is also confirmed that the impedance wall significantly changes the pressure-strain terms in the budget of normal stresses, which eventually leads to the redistribution of TKE.

The only previous work applying IBCs in a hypersonic flow is limited to transitional flows [Fedorov *et al.* (2003); Sousa *et al.* (2019)]. With this paper we start to explore the possibility of controlling a high-speed fully developed turbulent flow. We have found the range of resistance values that yield an effect on the flow and have characterized what type of structural changes turbulence is experiencing as a result. This is the first step toward something more meaningful which might require optimization strategies applied to a variable surface impedance.

ACKNOWLEDGMENTS

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TABLE III. Equivalent dimensional pore size d_{pore}, boundary layer thickness δ_{bl} (or channel half height) and characteristic resonant frequency f_{res} for current supersonic channel flow calculations and for two existing hypersonic flow cases.

Application	Mach	d_{pore}	δ_{bl}	f_{res}
Current Channel Flows Conditions	1.5	$\sim$100 μm	4.9 mm	228 kHz
	3.5	$\sim$100 μm	4.9 mm	531 kHz
Hypersonic Flow (T3-26 Tunnel) [19]	6.0	100 μm	1 mm	225–375 kHz
Hypersonic Flow (HEG Tunnel) [41]	7.4	20–100 μm	1 mm	300 kHz

APPENDIX A: ESTIMATION OF PORE SIZES IN HIGH-SPEED FLOWS

In Table III we analyze some realistic supersonic and hypersonic flow conditions, relating them to our permeable-wall channel calculations made dimensional by scaling the flow parameters to match HyShot II's combustor inlet at one working condition [39,40]. This results in dimensional values for effective pore diameters d_{pore} and resonating frequencies f_{res} of the impedance boundary condition that can be compared to realistic spatial and temporal scales.

The dimensional channel half height is chosen to be $\delta = 4.9$ mm, which can be interpreted as a reference boundary layer height. Varying the Mach number in the range $M = 1.5$–3.5 then results in a corresponding range of resonating frequencies $f_{\text{res}} = 228$–531kHz for the three-parameter IBCs used in our study. The frequencies are of the same order of magnitude as what naturally observed in the hypersonic flow cases mentioned in the table (T3-26 and HEG Tunnels). Assuming a wall porosity consistent with the adopted three-parameter model (i.e., an idealized Helmholtz resonator), the resulting resonating frequencies have been calculated via

$$f = \frac{a}{2\pi}\sqrt{\frac{A}{V_0\,\ell_{\text{eq}}}}, \tag{A1}$$

where a is the speed of sound, $A \sim d_{\text{pore}}^2$ the orifice's opening area, V_0 is the resonating cavity volume and ℓ_{eq} is the effective neck length. In the Hyshot II setup, the wall temperature is chosen as $T_w = 300K$ and the resulting speed of sound at the wall is $a_w \approx 347$ m/s. We assume V_0 and ℓ_{eq} share the common characteristic length scale ℓ_c given the typical structure of porous walls used in high-speed applications, resulting in $V_0 \sim \ell_c^3$ and $\ell_{\text{eq}} \sim \ell_c$. Choosing a pore diameter of $d_{\text{pore}} \sim 100\,\mu$m (and hence $\ell_c \sim 100\,\mu$m), this results in the frequency range relevant to the first two cases of Table III. Another point is that values considered for the resistance (to be interpreted as the inverse of the wall permeability) are never lower than 0.5, further supporting the assumption of small pore size.

The small pore sizes reported here entail typical pore-size-to-Stokes-boundary-layer-thickness ratios of less than 0.1. This means that the wave-induced flow in the porous media is heavily viscously dominated, and the acoustic waves are strongly attenuated in the depth of the porous layer making nonlinear effects and slip-flow effects at the surface unlikely to be relevant.

With this pore size, one can do a back-of-the-envelope calculation about how many grid points would be required if the pores are fully resolved. Take the case $M_b = 3.50$ as an example whose pore size is about $1/49$ of the channel half height δ. Further assume at least five grid points must be used to resolve the pore (in each parallel direction, i.e., x and z) and the same pore spacing, then a well-resolved simulation would require about 5 times of the points in x and 2 times of the points in z as compared to the current simulation, not including the points required to resolve the interior of cavities and pores. The current calculations are computationally intensive due to the nonzero wall-normal velocities caused by porosity near the IBC, creating rather constraining convective CFL conditions. For example, the current grid size for the case $M_b = 3.50$, $R = 0.50$ is $340 \times 300 \times 300$ and one flow-through requires 137143 CPU hours. An equivalent pore-resolved simulation would at least need 0.3 billions of grid point at such high speed and definitely way more CPU hours.

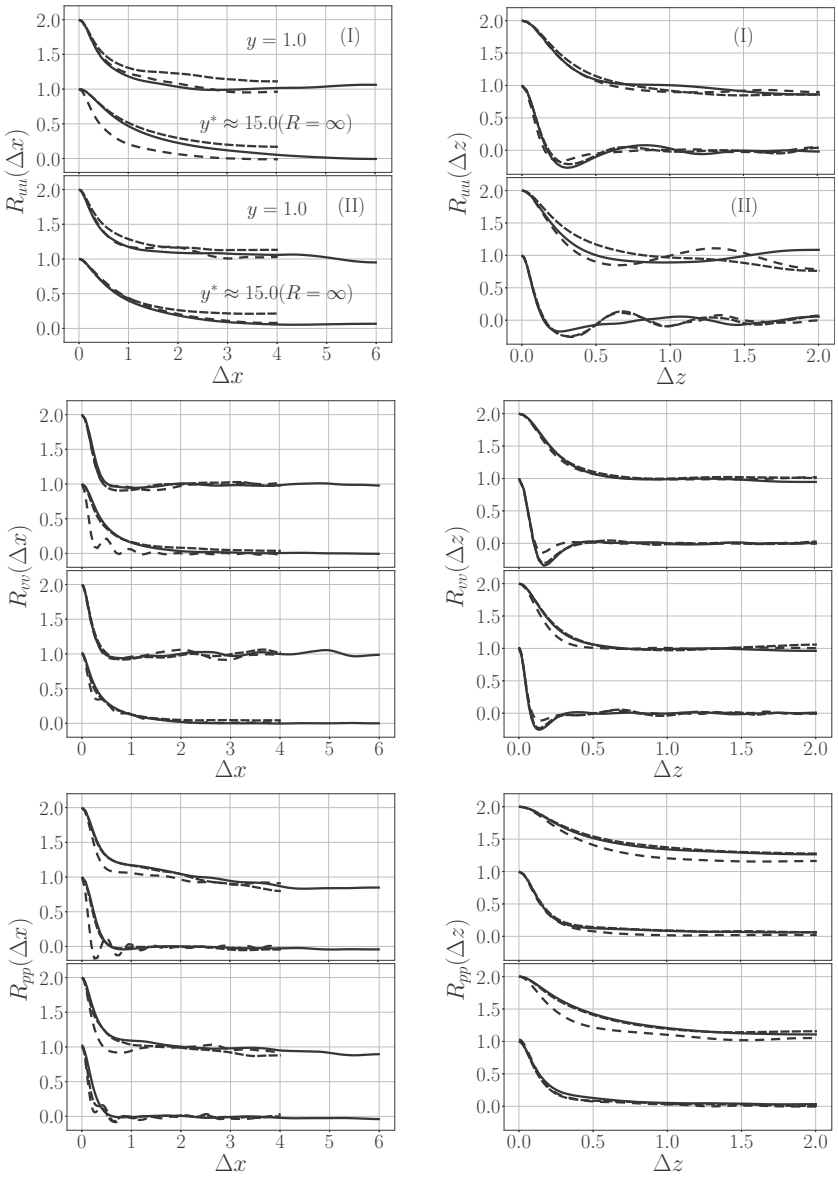

FIG. 9. Two-point correlations in streamwise direction (left column) and spanwise direction (right column) for all cases presented in this work. Variables being evaluated are u, v and p. Two locations are picked: $y^* \approx 15 (R = \infty)$ and $y = 1$ (channel center). Note that for impermeable wall cases ($R = \infty$), the domains are longer in streamwise direction as compared to permeable wall cases.

APPENDIX B: TWO-POINT CORRELATION OF CHANNEL FLOWS

Here the two-point correlations of all cases are presented in Fig. 9, which show sufficient domain length in periodic directions.

APPENDIX C: GRID SENSITIVITY STUDY

The grid sensitivity study is conducted with case $M_b = 3.50, R = 0.50$, which exhibits most energetic flow response from permeable wall. Three grids are tested. First and second order mean

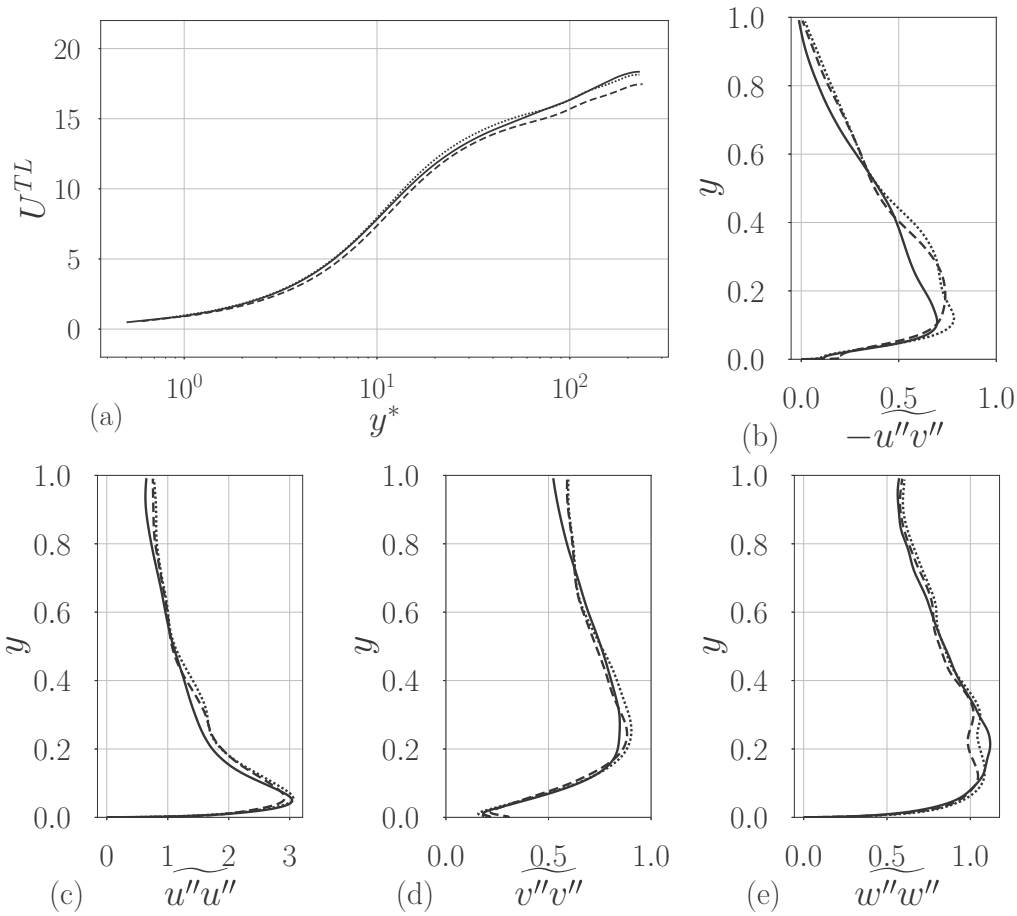

FIG. 10. First and second order statistics of case on different grids of case B2 ($M_b = 3.50$, $R = 0.50$). Grid sizes presented are: $200 \times 200 \times 180$ (—), $270 \times 256 \times 240$ (—-),$340 \times 300 \times 300$ (—). The quantities shown are (a) mean streamwise velocity in TL-transformation, (b) normalized Reynolds shear stress, (c)–(e) normalized normal stresses. The normalization of the stresses follows the convention $\overline{\rho u_i'' u_j''}/\tau_w$.

flow statistics are plotted in Fig. 10. The mean flow statistics are found to be more sensitive to the grid than impermeable wall cases. Meanwhile the profiles in the outer layer converge much slower than those in the inner layer. The results show decent convergence in the near-wall region, which is the main focus of the current study.

APPENDIX D: TRANSPORT EQUATION OF REYNOLDS STRESS

The derivation of transport equation of Reynolds stress is slightly tedious and is thus omitted here, readers can consult textbooks such as [30]. The result is given as follows with $R_{ij} = \widetilde{u_i'' u_j''}$ denoting the Reynolds stress tensor:

$$\frac{\partial \overline{\rho} R_{ij}}{\partial t} + \frac{\partial \overline{\rho} \widetilde{u}_k R_{ij}}{\partial x_k} = -R_{ik}\frac{\partial \widetilde{u}_i}{\partial x_k} - R_{jk}\frac{\partial \widetilde{u}_j}{\partial x_k} - \epsilon_{ij} + \Pi_{ij} + \frac{\partial}{\partial x_k}[\overline{\tau_{kj} u_i'' + \tau_{ki} u_j''} - C_{ijk}] - \overline{u}_i''\frac{\partial P}{\partial x_j} - \overline{u}_j''\frac{\partial P}{\partial x_i}, \tag{D1}$$

where

$$\epsilon_{ij} = \overline{\tau_{kj}\frac{\partial u_i''}{\partial x_k} + \tau_{ki}\frac{\partial u_j''}{\partial x_k}}, \quad \Pi_{ij} = \overline{p'\left(\frac{\partial u_i''}{\partial x_j} + \frac{\partial u_j''}{\partial x_i}\right)}, \quad C_{ijk} = \overline{\rho u_i'' u_j'' u_k''} + \overline{p' u_i''}\delta_{jk} + \overline{p' u_j''}\delta_{ik}. \quad \text{(D2)}$$

Equations for the normal stresses can be obtained by simply setting $i = j$.

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